

Critical Design Review (CDR) – Group 3 – Eyobed, Asrael, Mohammed, Issac

Objective

The objective of this project is to design and prototype a wearable medication reminder wristband that provides a clear vibration alert at a scheduled time and allows the user to dismiss the reminder with a single button press. The device is intended to operate independently of a phone by using an onboard real-time clock (RTC) and microcontroller to keep time and trigger alerts.

The current prototype uses an Arduino Nano-class board, a DS3231 RTC module, a vibration motor driven through a transistor, and a pushbutton mounted on perfboard. For this CDR, the focus is on verifying that the core alert pathway (timekeeping → vibration alert → user dismissal) works reliably enough to justify finishing the build as a wearable.

Current Overview

During operation, the microcontroller initializes the RTC, then repeatedly reads the current time over I2C. When the RTC time matches the programmed alarm time, the firmware drives D5 high to switch on the transistor and vibrate the motor. The alert continues until the button on D3 is pressed; the button press is detected in software and used to stop the motor.

Prototype Behavior

Recent bench tests show that:

- The DS3231 now returns valid date and time values, confirming working I2C wiring and communication.
- The alarm logic compares RTC time against a scheduled alarm time and starts the vibration motor at the correct second.
- The momentary pushbutton on D3 is handled in software so a single press reliably dismisses the alert.

These tests demonstrate that the core functional loop from timekeeping to alert to user acknowledgment is working in the current prototype.

Risks

Technical Risks

- **RTC without backup battery:** The DS3231 module is currently operated without a coin-cell backup battery, so it loses the correct time whenever power is removed.
- **No power-management yet:** The current prototype is powered like a bench-top Arduino project. Will be using a lithium battery for the final though.

Usability / Human-Factors Risks

- **Alert strength and duration:** The vibration amplitude and duration have only been tuned informally. There has been no user testing to confirm that the alert is strong enough to notice but not so strong or long that it becomes annoying.

Design Changes Since PDR

Since the Preliminary Design Review, the following key changes and clarifications have been made:

1. **Confirmed button behavior and software handling**

The initial concept assumed a latching switch to leave the device in a “dismissed” state. The actual hardware uses a pushbutton. The firmware has been implemented so that a press during an active alarm reliably turns the motor off, preserving the intended press-once-to-dismiss user experience.

2. **Corrected RTC wiring and validated readings**

Early experiments showed incorrect or nonsensical time values from the DS3231, traced to incorrect or intermittent SDA/SCL wiring. The RTC is now wired with SDA to A4 and SCL to A5, and the code successfully reads plausible date and time values from the module.

3. **Integrated full alarm path into a single sketch**

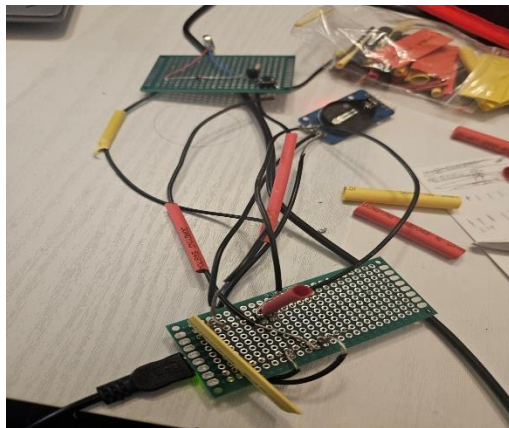
Separate test sketches for the motor, button, and RTC have been consolidated into one program that sets the RTC time, continuously monitors time, triggers the vibration alert at a scheduled time, and turns the motor off when the button is pressed.

4. **Progressed from loose wiring to perfboard layout**

Components have been moved from a loose breadboard configuration to perfboard, which better approximates a compact wearable and reduces accidental disconnections.

These changes increase confidence in the design concept and reduce the risk that basic hardware issues will block remaining work.

Prototype Picture



To-Do / Doing / Done

Done

- Implemented and tested the complete alert pathway:

- Motor is controlled from the microcontroller through a transistor and flyback diode.
- button on D3 dismisses an active alert using debounce and toggle logic in software.
- DS3231 RTC on A4/A5 returns valid time readings and can be set from the sketch at startup.

Doing

- **Stabilizing prototype integration**
 - Cleaning and securing perfboard wiring so that I2C, power, and motor connections are robust to handling and movement.
 - Selecting and testing a small battery solution for powering the system away from USB.
 - Deciding whether or not to implement the message sending/pill storage
- **Refining firmware for repeated daily use**
 - Adjusting the code so that once the RTC has a backup battery, time is only set when needed rather than on every reset.

To-Do

- **Add RTC backup battery and verify retention**
 - Install the appropriate coin-cell battery on the DS3231.
 - Confirm that the RTC keeps accurate time across power cycles and overnight, and that alarms still trigger correctly after long intervals.
- **Design and build a wearable enclosure**
 - Create or adapt a wristband enclosure to hold the perfboard, motor, and battery securely.
 - Ensure the vibration motor has good contact with the wrist and the button is easy and comfortable to press.
- **Finalize documentation and demonstration**
 - Prepare and record a short demo showing the device triggering at a scheduled time and being dismissed by the user.

Conclusion

The current design of the medication reminder wristband successfully demonstrates the critical functional loop of the system: maintaining time with a DS3231 RTC, triggering a vibration alert at a scheduled time, and allowing the user to dismiss that alert with a single button press. Remaining risks include the lack of a backup battery on the RTC, wiring, and enclosure design, but these can be addressed without fundamentally changing the architecture.